Engineering Guideline: Concrete Curing for Structural Columns

Document Number: ENG-GL-012, Revision B (Dummy/Fictional content — for demo purposes only)

1. Scope

Minimum curing duration and methods for cast-in-place structural columns on site. Curing durations updated below following validation of a new high-early-strength admixture.

2. Curing Requirements

- Minimum wet curing duration is 5 days for standard Portland cement, 7 days when fly ash content exceeds 20% (reduced from Revision A following validated use of a high-early-strength admixture).
- Curing compound must be applied within 2 hours of final finishing if wet curing is impractical.
- Ambient temperature during curing must stay between 10C and 35C; outside this range requires engineer approval.
- Formwork for columns may be stripped after 24 hours but curing must continue by alternate method.
- Compressive strength test cylinders must be cast from the same batch and cured under matching conditions.

3. Non-Conformance

Any deviation from this guideline must be documented in a Non-Conformance Report (NCR) and approved by the Structural Engineer of Record before concrete placement proceeds on subsequent elements.

4. Revision History

Revision B supersedes Revision A. Minimum curing durations reduced per updated admixture validation testing described in Section 1.